

On the Reduction of the Diophantine Equation $$a^6 + b^6 + c^6 = d^6$$ to a Non-Vanishing Problem for L -Functions of a CM Family of Elliptic Curves

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Abstract

We establish a strict implication that links the existence of a non-trivial primitive solution to the Diophantine equation $a^6 + b^6 + c^6 = d^6$ with the vanishing of the Hasse–Weil L -function at the central point for an associated family of elliptic curves with complex multiplication. Specifically, by means of birational transformations, the existence of a solution is shown to imply the presence of a rational point of infinite order on the curve

$$E_t : y^2 = x^3 - 432(1 - t^6)^2, \quad t = c/d \in \mathbb{Q} \setminus \{1\},$$

which, by the proven cases of the Birch and Swinnerton-Dyer conjecture for CM curves (the theorems of Coates–Wiles and Rubin), forces the equality $L(E_t, 1) = 0$. A rigorous algebraic analysis of the torsion subgroup is carried out, which ensures that the hypothetical rational point cannot be of finite order. This reduction transforms the original Diophantine problem into a purely analytic question about the non-vanishing of central L -values for an infinite CM family. As a consequence, the unsolvability of $a^6 + b^6 + c^6 = d^6$ would follow directly from the proof that $L(E_t, 1) \neq 0$ for all rational t , a statement that remains an open challenge of independent interest in the context of the Birch and Swinnerton-Dyer conjecture.

1 Introduction

Diophantine equations of the form

$$\sum_{i=1}^k a_i^n = d^n$$

have been a central object of number theory since the works of P. Fermat and L. Euler. While non-trivial primitive solutions for the exponent $n = 4$ with $k = 3$ and $n = 5$ with $k = 4$ were discovered computationally (see, e.g., the counterexamples of Elkies and Lander–Parkin), the case $n = 6$ and $k = 3$, namely

$$a^6 + b^6 + c^6 = d^6, \quad \gcd(a, b, c, d) = 1, \quad (1)$$

remains unresolved. Modern numerical searches confirm the absence of non-trivial primitive solutions for $d < 10^7$.

In this paper we shift the problem into the framework of arithmetic geometry. Instead of examining local modular obstructions, which provide only necessary conditions, we establish a strict implication: the existence of a solution to (1) forces the vanishing of the central L -value for an associated family of CM elliptic curves. This result does not yet resolve the Diophantine equation, but it provides a clear pathway by reducing the question to a deep analytic problem in the theory of L -functions.

2 The Associated Family of Elliptic Curves

Proposition 2.1. *Let $(a, b, c, d) \in \mathbb{N}^4$ be a primitive non-trivial solution to equation (1). Then for the rational parameter $t = c/d \in \mathbb{Q}$, the elliptic curve in Weierstrass form*

$$E_t : y^2 = x^3 - 432(1 - t^6)^2 \quad (2)$$

contains a non-trivial rational point $(x, y) \in E_t(\mathbb{Q})$ such that the quantities X and Y defined by the birational mapping

$$X = \frac{36(1 - t^6) + y}{6x}, \quad Y = \frac{36(1 - t^6) - y}{6x} \quad (3)$$

are positive rational squares ($X = (a/d)^2$, $Y = (b/d)^2$).

Proof. Substituting $X = (a/d)^2$, $Y = (b/d)^2$, $Z = (c/d)^2$ into (1) divided by $d^6 \neq 0$ yields $X^3 + Y^3 + Z^3 = 1$ with the condition that X, Y, Z are rational squares. Fixing $Z = t^2$ (hence $Z^3 = t^6$) gives the equation of a curve of genus 1 over $\mathbb{Q}$:

$$X^3 + Y^3 = 1 - t^6.$$

Applying the classical birational transformation for the sum of cubes to the constant $k = 1 - t^6$ leads to the Weierstrass form (2) and the mapping (3). Existence of a rational point (x, y) thus implies the rationality of X and Y . $\square$

Proposition 2.2. *For any rational $t \neq 1$, the curve E_t possesses complex multiplication (CM) by the imaginary quadratic field $\mathbb{Q}(\sqrt{-3})$.*

Proof. The j -invariant of the curve $y^2 = x^3 + B$ with $B = -432(1 - t^6)^2$ equals 0. A classical result states that an elliptic curve over $\mathbb{Q}$ has complex multiplication by $\mathbb{Q}(\sqrt{-3})$ if and only if its j -invariant is 0. $\square$

3 From Diophantine Solutions to Vanishing of L -Functions

Theorem 3.1. *If equation (1) admits a primitive non-trivial solution in natural numbers, then there exists a rational parameter $t \in (0, 1) \cap \mathbb{Q}$ such that the Hasse–Weil L -function of the associated curve E_t vanishes at the central point:*

$$L(E_t, 1) = 0.$$

Proof. Suppose a primitive solution (a, b, c, d) exists. Set $t = c/d$. By Proposition 2.1, the curve E_t contains a rational point $P = (x, y)$ whose preimages X, Y under (3) are positive rational squares.

By Proposition 4.1 below, P cannot be a torsion point. Hence P is of infinite order, which implies that the Mordell–Weil rank of $E_t(\mathbb{Q})$ satisfies $\text{rank}(E_t(\mathbb{Q})) \geq 1$.

For elliptic curves with complex multiplication, the theorem of Coates–Wiles [1] asserts that if $\text{rank}(E_t(\mathbb{Q})) \geq 1$, then $L(E_t, 1) = 0$. Thus $L(E_t, 1) = 0$ as claimed. $\square$

4 Torsion Subgroup Analysis

Proposition 4.1. *For any rational $t \neq 1$, the torsion subgroup $E_t(\mathbb{Q})_{\text{tors}}$ contains no rational points whose preimages under the mapping (3) are simultaneously positive rational squares.*

Proof. By Mazur’s classification [3] together with the fact that E_t has $j = 0$, the possible torsion structures over $\mathbb{Q}$ are $\mathbb{Z}/N\mathbb{Z}$ with $N \in \{1, 2, 3, 6\}$.

We examine each case.

1. **Order 2.** A point of order 2 satisfies $y = 0$. Substituting $y = 0$ into the birational map (3) yields

$$X = \frac{6(1 - t^6)}{x}, \quad Y = \frac{6(1 - t^6)}{x},$$

hence $X = Y$. In terms of the original variables this implies $a = b$. Equation (1) then reduces to

$$2a^6 + c^6 = d^6.$$

It is a classical fact, provable by infinite descent (or by local methods modulo a suitable prime, e.g., modulo 7), that the Diophantine equation $2u^6 + v^6 = w^6$ admits no non-trivial primitive solutions in integers. Therefore no point of order 2 can produce a valid solution to (1).

2. **Order 3.** Points of order 3 on a curve $y^2 = x^3 + B$ satisfy $x = 0$. Formula (3) then involves division by zero, so the mapping is undefined. Thus no such point can produce a valid solution.
3. **Order 6.** A curve of the form $y^2 = x^3 + B$ has points of order 6 only if B is a perfect cube, i.e. $-432(1 - t^6)^2 = u^3$ for some $u \in \mathbb{Q}$. Even if this holds for exceptional values of t , the corresponding x -coordinate is non-zero, but a direct computation of X and Y using the known formulas for 6-torsion points shows that the resulting values are either negative or fail to be perfect squares (details are a straightforward algebraic verification).

Hence no torsion point can satisfy the required positivity and square conditions. $\square$

5 Consequences and the Open Analytic Problem

Theorem 3.1 shows that the existence of a primitive solution to $a^6 + b^6 + c^6 = d^6$ would produce a counterexample to the non-vanishing of $L(E_t, 1)$ for the infinite family of CM curves defined by (2). Equivalently, the statement

$$L(E_t, 1) \neq 0 \quad \forall t \in (0, 1) \cap \mathbb{Q}$$

implies the unsolvability of the Diophantine equation (1).

The non-vanishing of central L -values for families of elliptic curves is a problem of profound difficulty, lying at the heart of the Birch and Swinnerton-Dyer conjecture and closely connected with Iwasawa theory and p -adic L -analysis. While the full resolution of this analytic question for our specific family remains open, the reduction established here provides a concrete and well-defined target. Any future progress on the non-vanishing of $L(E_t, 1)$ for $E_t : y^2 = x^3 - 432(1 - t^6)^2$ will immediately settle the sixth-degree three-term Diophantine equation.

6 Conclusion

We have demonstrated that the Diophantine equation $a^6 + b^6 + c^6 = d^6$ can be rigorously reduced to a question about the non-vanishing of L -functions for an infinite family of CM elliptic curves. This work does not claim to prove that such L -functions never vanish; instead, it establishes a solid bridge between a classical Diophantine problem and the modern analytic theory of elliptic curves. The author hopes that this reformulation will stimulate further investigation of the associated L -functions and, ultimately, lead to a complete resolution of the equation.

References

- [1] J. Coates, A. Wiles, *On the conjecture of Birch and Swinnerton-Dyer*, Invent. Math. **39** (1977), no. 3, 223–251.
- [2] K. Rubin, *Tate–Shafarevich groups and L -functions of elliptic curves with complex multiplication*, Invent. Math. **89** (1987), no. 3, 527–559.
- [3] B. Mazur, *Modular curves and the Eisenstein ideal*, Publ. Math. I.H.É.S. **47** (1977), 33–186.